



Suspended animation

Doctors have placed humans in suspended animation for the first time as part of a trial in the US. <https://digit.in/dec19-61>



Old bird, new song

Nightingales practise new songs in the winter to impress potential mates in spring. <https://digit.in/dec19-62>

In 2019, researchers at UC Berkeley have developed a process for manufacturing sweat sensors on rolls of flexible plastic substrate, similar to a filmstrip of an analog camera. The sensors are embedded onto the plastic in a manner similar to the way a printing press imprints text on a newspaper. Inside the sensors are spiralling microfluidic tubes, that can derive information on the rate at which the wearer is sweating. The tubes also have tiny chemical sensors that can measure the concentrations of potassium, sodium and metabolites, a class of molecules that inter-



Credit: Sam Esham/UC Berkeley

The wearable sweat sensor developed by the team from UC Berkeley and Stanford

act with enzymes. The process allows for production of scale, which means large volumes at low cost. In initial tests, the trackers could understand when the liquid loss of the volunteer testers was too much during intense exercise sessions.

In November this year, a team at Caltech led by professor Wei Gao published a paper describing an even more sensitive sweat sensor in *Nature Biotechnology*. The previous sensors could monitor concentrations of electrolytes, lactate and metabolites, all of which occur in high concentrations in sweat. The sensor developed by Gao's team can detect molecules in much lower concentrations, increasing the range of conditions that can be detected by sweat sensors. The sensor developed in Caltech can be used to track cardiovascular, liver or kidney diseases as well as neuropsychiatric conditions and eating disorders. This

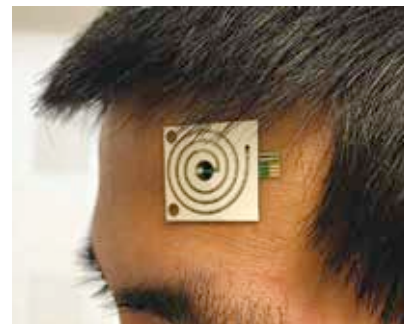
is accomplished by the additional monitoring of uric acid and an amino acid called tyrosine. Gao explains, "Such wearable sweat sensors have the potential to rapidly, continuously, and noninvasively capture changes in health at molecular levels. They could enable personalised monitoring, early diagnosis, and timely intervention." Gao's team created the microfluidic channels in plastic film to capture the sweat using a carbon dioxide laser, and used the same device to craft the more sensitive graphene sensors. The performance of the sensors was checked by comparing the measurements with those derived from traditional blood tests. The tests showed that the sweat sensors were doing their jobs.

THE BENEFITS

Over 400 million people around the world suffer from diabetes, and hundreds of millions more are prediabetic. Sweat sensors hold the promise of helping them tremendously, as one of the metabolites that sweat sensors can measure is glucose. Traditionally, home tests for measuring glucose levels for diabetic patients involved devices that drew blood with a fingerprick. Sweat sensors can do away with such devices, and additionally provide real time tracking of blood glucose levels. Diabetic patients have to stay within a narrow healthy range by ensuring that their blood sugar levels do not spike by too much, or suddenly fall, both of which can have serious consequences. By pairing such a sensor with a smartphone app,

a person can take the countermeasures well in advance of a situation.

Apart from finding out if the patient has any problems, in particular cases, dedicated sweat sensors can make sure that the patient is following the recommended treatment,



Credit: Uliam Mossley

A head mounted sweat sensor developed by UC Berkeley researchers

and track the progress of recovery. Students from Johns Hopkins University have developed such a sensor for patients suffering from cystic fibrosis, a genetic condition linked to respiratory problems. Patients suffering from the disease have a high level of chloride in their sweat. A new class of drugs to treat the problem by addressing the protein problems that cause symptoms of the disease. The treatment regime also reduces the concentration of chloride in the sweat. For patients suffering from cystic fibrosis, sweat sensors offers a way to track the progress of treatment and make the appropriate adjustments if necessary. Healthcare in general stands to benefit from continuous, real time monitoring of the patient's health. In the future, these sensors could find their way into smartclothes, fitness trackers or smartwatches, and not require a dedicated band or a sticker just for sweat monitoring. While a fitness tracker today can monitor activity levels and heart rate, a sweat sensor can provide so much more information for healthcare professionals. The technology promises to be particularly helpful to those living in remote and rural areas, away from easy and quick access to healthcare. 



Credit: Caltech

The flexible sensor developed by Wei Gao and team